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RESEARCH STUDY ON ADVANCED DRIVER- ASSISTANCE SYSTEMS (ADAS) - GLOBAL & INDIAN AUTOMOTIVE SCENARIO

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ABSTRACT

In Future, Almost all vehicles will be fully or semi-automated, in that scenario the ADAS systems will be the most crucial part of Automation. The goal of the ADAS system is to control injuries by reducing the number of car accidents. With the help of modern algorithms and tools it supports real-time multimedia, vision co-processing, and sensor fusion subsystems. The features of ADAS system includes the following Pedestrian detection/avoidance, Lane departure warning/assistance, traffic light Identification, Automatic emergency braking, Blind spot detection. This Journal explains the Growth and development of the Advanced Driver Assistance System(ADAS) in India compared to abroad. It also explains various technologies involved in ADAS.

Keywords: Advanced Driver Assistance System, Latest Interface Standard Vision-Based Algorithms, Sensor Fusion Subsystem, Technologies Involved.

I. INTRODUCTION

Advanced driver Assistance systems (ADAS) has many electronic as well as digital instruments that helps driver in reducing errors and parking functions etc. Its major goal is to prevent accidents by reducing human errors. Through the advanced interface technologies, ADAS increases the safety of car and the environment. ADAS use automated equipment such as sensors and cameras to detect obstacles or driver errors and reacts to them instantly. Advanced driver assistance systems (ADAS) are becoming more and more common in vehicles. Some of us may be driving our car with ADAS without even realizing it. The technology itself is based on information from a camera or other sensors that can see further down the road than a human being—usually about 200 meters in front of the vehicle. That distance is much longer than what our eyes can see—about 80 meters—and so this extra information from ADAS provides us with better awareness of what lies ahead. ADAS systems are also designed to make driving more fun. They can make it easier to parallel park, or they can help you to follow a particular route that has been programmed into the system. And, if you are an experienced driver, the technology can even help you avoid major accidents by helping you detect potential hazards in time to take evasive action. Advanced stop start systems are another kind of ADAS that have become common on new vehicles. These systems enable a vehicle to start quickly when stopped at traffic signals. As most road accidents occur due to human error, ADAS are developed to automate, adapt, and enhance vehicle technology for safety and better driving. ADAS reduces road fatalities by minimizing human mistakes due to carelessness. Safety features are designed to prevent collision by offering functions that alert the driver about the upcoming danger, implementing preventive measures and taking control of the vehicle if needed. Adaptive features enhances lighting, provide adaptive control , assist in avoiding collisions, incorporate satellite navigation and traffic warnings, alert drivers to possible obstacles, assist in lane departure and lane centering, provide navigational assistance through smartphones, and supply other features. The development of ADAS technology in the upcoming days will be huge and some chart showcasing the development is shown below

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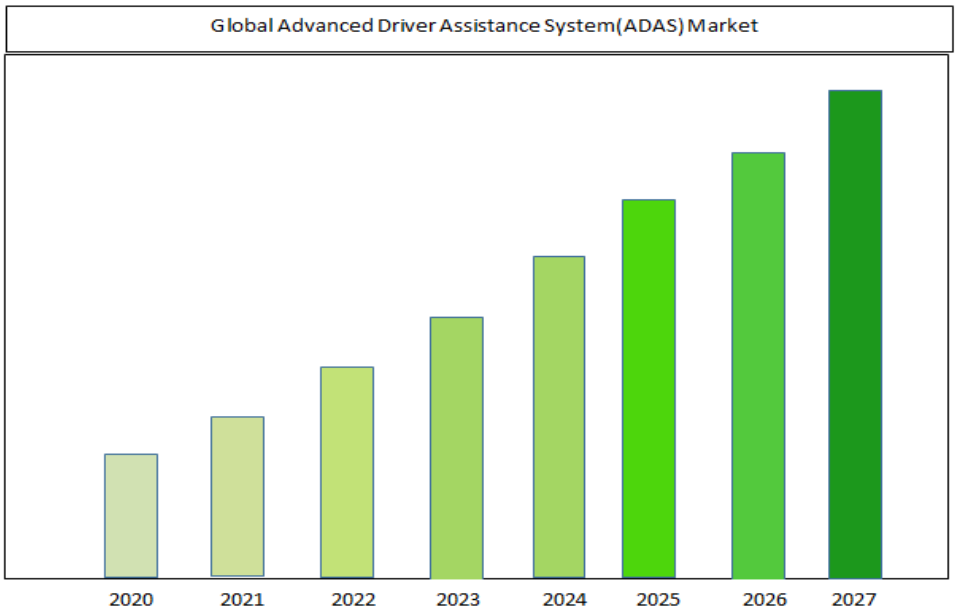
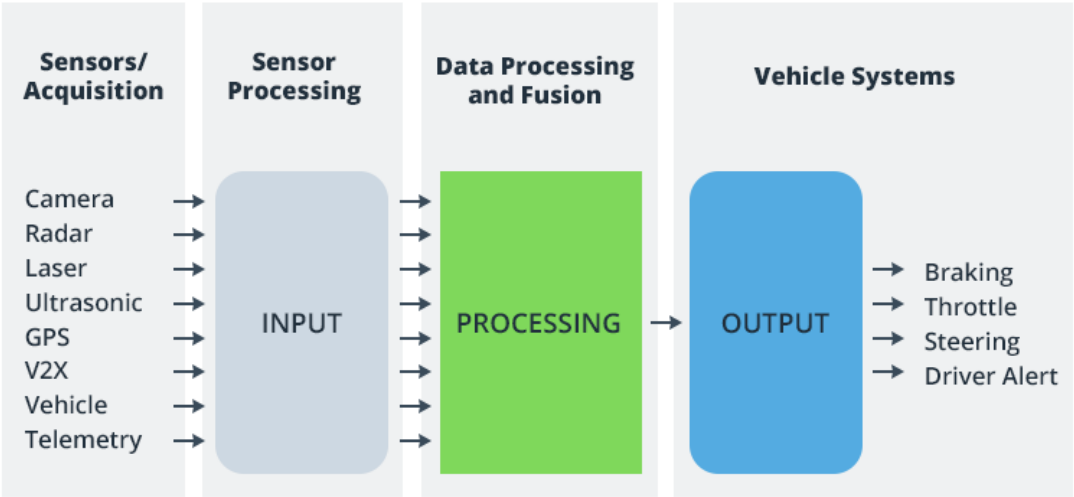


There are five important things that helps ADAS to assist autonomous driving. They are Theses are grouped into following three types

- Data acquisition and processing
- Data fusion and decision-making
- Taking action

II. DESIGN OF ADAS SYSTEM



Source: embedded-vision.com

III. HISTORY

The first used ADAS system is the ABS(Anti lock Braking System).It is used in 1970's. The ABS system was first developed for 1920s era aircraft to prevent skidding at the time of landing, then Robert Bosch, in joint development with Mercedes-Benz used ABS system on automobiles. The load sensing proportioning valve used in the mid-1960s is also one of the assisting systems. Proportioning valves were installed on pickup trucks to minimize vehicle spin (swapping ends) during hard braking on wet roads. It helps to +control the brake fluid pressure from the master cylinder and prevents early locking of the rear wheels.

IV. LEVELS OF ADAS SYSTEMS

There are Six levels in ADAS system that enhances the driver's experience and comfort. These levels are set by the society of Automotive Engineers across the globe. They are classified as follows:

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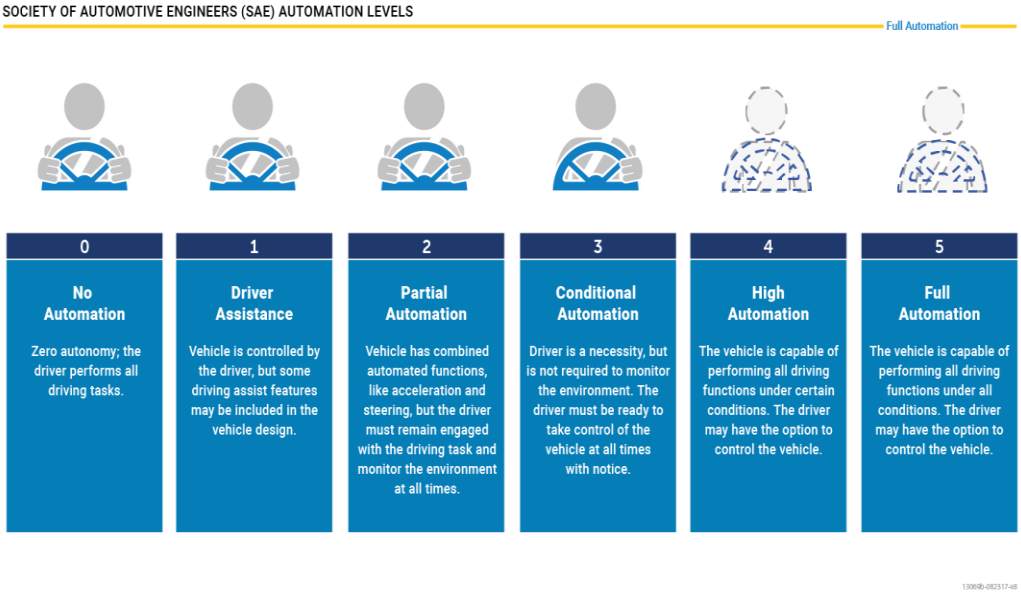

Source: accoladetechnology.com

Level 0(No Automation)	All Things are done only by driver
Level 1(Driver Assistance)	The Advanced driver assistance system on the vehicle can assist the driver with either steering or braking/accelerating sometimes, but not both simultaneously.
Level 2(Partial Automation)	ADAS itself controls both steering and braking/accelerating simultaneously under some situations . The Driver must continuously watch the road and surroundings .
Level 3(Conditional Automation)	The control is done Automatically but in critical situations the driver should take back control when insisted by the system.
Level 4(High Automation)	The ADAS can drive the car by itself as well as monitor the environment and needs the driver's handling only in some situation.
Level 5(Full Automation)	All the tasks are fully done by system. The humans are just passengers.

These levels of Advanced driver Assistance helps in making the driving experience more comfortable and safe. Not only it helps the driver but also helps in preventing many road accidents especially in a country like INDIA where road accidents are more in number. It helps in preventing the human errors in driving and assists them with advanced technologies.

V. REDUCTION IN ACCIDENT RATES AFTER ADAS SYSTEM

Forward collision warning system with auto brake:

Front to rear crashes = -27%

Front to rear crashes with injuries= -20%

Claim rate for damage to other vehicles= -9%

Claim rate for injuries to people in other vehicles= -17%

Lane Departure warning:

Single vehicle sideswipe and head on crashes= -11%

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Injured crashes = -21%

Blind spot deduction:

Lane change crashes= -14%

Lane change crashes with injuries= -23%

Damage to other vehicles= -7%

Rate of injuries for people in other vehicle= -9%

Rear automatic braking:

Backing crashes= -78%

Damage to other vehicles= -28%

Damage to own vehicle= -10%

VI. LEVEL OF ADAS SYSTEM USED IN INDIA

As per the current scenario, the level 2 of ADAS system is used in INDIA. One of the Car's that is providing this feature currently is the XUV 700 at a reasonable cost. The pricing of this XUV ranges between INR 18 - 25 lakh depending upon the variant.

The Level 2 ADAS system used by XUV 700 includes the following

1. **FCW/AEB:** Front Collision Warning and Auto Emergency Braking (AEB). These uses the data from the front sensors and cameras to detect vehicles coming to a stop in front of our vehicle or pedestrians coming in the way. The FCW system warns the driver to apply the brakes to avoid a collision, and if the driver doesn't react within a certain time, the AEB system automatically engages the brakes and stops the vehicle. In minimum speeds, this system can avoid a collision, whereas in higher speeds, it prevents the severity of the collision.
2. **LKA:** Lane Keep Assist (LKA) helps in keeping the vehicle centred in its lane. It uses lane markings to maintain the center and can also detect cars on the side of a vehicle. Similar to AEB it first warns and then takes control to maintain the lane.
3. **BDS:** Blind-spot Detection System (BDS) helps us when cruising on a highway. It has sensors or cameras at the side and rear of a vehicle which helps us to detect vehicles that is in the blind spot of the side and center of the vehicle's mirror.
4. **ACC:** Adaptive Cruise Control (ACC) helps in maintaining the speed of the vehicle in a steady state and also with respect with others in front. This system maintains speed by increasing or decreasing the speed according to the vehicle in front of the lane.
5. **Rear Cross-Traffic Alert:** This system helps us in reversing the vehicle safely. The side sensors or cameras can detect vehicles approaching from either side, warn the driver, and even take corrective actions by applying the brakes.
6. **Drowsy driver detection system:** It warns the driver if they are drowsy and alerts them by making sound. It prevents accidents due to continuous work.

VII. HIGHEST LEVEL OF ADAS SYSTEM IN FOREIGN COUNTRIES AND ITS COMPARISON

The Tesla uses autopilot (Level 3) feature in its cars. The car can manually operate in self-driving mode. But until legislation and infrastructure evolves, they can only do so within a limited area (usually an urban environment where top speeds reach an average of 30 mph). This is known as geofencing.

FEATURES OF ADAS SYSTEM	FEATURES CURRENTLY USED IN INDIAN ROADS (XUV700)	FEATURES USED ABROAD ROAD CONDITIONS (TESLA MODEL S)
Adaptive Cruise Control (ACC)	Yes	Yes
Glare-Free High Beam	Yes	Yes

and Pixel Light		
Adaptive Light Control	Yes	Yes
Automatic Parking	No	Yes
Navigation System	Yes	Yes
Night Vision	No	Yes
Blind Spot Monitoring	Yes	Yes
Automatic Emergency Braking	Yes	Yes
Crosswind Stabilization	No	Yes
Driver Monitoring System	Yes	Yes

VIII. MAJOR DISADVANTAGES IN OF ADAS

The ADAS is a complex system that involves huge amount of data transfer, analyzation and storage, so that the stored data helps in assisting the driver to prevent accidents due to human error. This system needs precise calculation and reaction as it involves live data collection in the vehicle movement. There are various problems involved in this system. Some of them are:

- Difficult to work in extreme climatic conditions
- The Automatic braking stops the vehicle by itself during short turns
- Processing huge amount of data is difficult
- Includes many sensors and cameras that costs more
- Service and Availability is difficult
- Needs Driver in critical situations (Example: traffic)
- Mostly useful only in highways

IX. CONCLUSION

The ADAS system is one of the crucial emerging feature in forthcoming cars. Thus the Effects of Advanced Driver Assistance system in the upcoming modern world is shown above. The difference between the ADAS systems used in India and Abroad is explained and its growth rate over the years is mentioned in this study paper

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